

Architecture Selection: We chose the Layered Architecture for the Personalized Meal & Diet Subscription Manager (Plate Pilot) System.

Architectural Choice: Layered Architecture (organized into Presentation, Business Logic, and Data layers).

Two Reasons:

1. **Clear Separation of Concerns:** This architecture separates the complex user interfaces (the Subscriber's mobile app and the Dietitian's dashboard) from the core backend nutritional mathematics, making both easier to maintain.
2. **Centralized Rule Management:** By routing all requests through a central business layer, the system can reliably enforce medical and dietary constraints (allergen checks, macro recalculations) in one place without duplicating code across different user portals.

Security Advantage: The system must encrypt patient metabolic data and restrict access (NFR-002). The Layered Architecture addresses this by isolating the Database Component at the bottom layer. The User Interface Component cannot query the database directly; it must pass through the business layer, which acts as a gatekeeper to ensure only authorized subscribers or assigned dietitians can access specific health records.

Performance Benefit: The system must process schedule modifications in under 5 seconds (NFR-001). Unlike Microservices, which introduce network latency and complex data synchronization, a Layered Architecture allows the Meal Planning Engine Component to communicate directly and instantly with the local Database Component for extremely fast data persistence and retrieval, easily meeting the latency target.

2. UML Component Diagram

